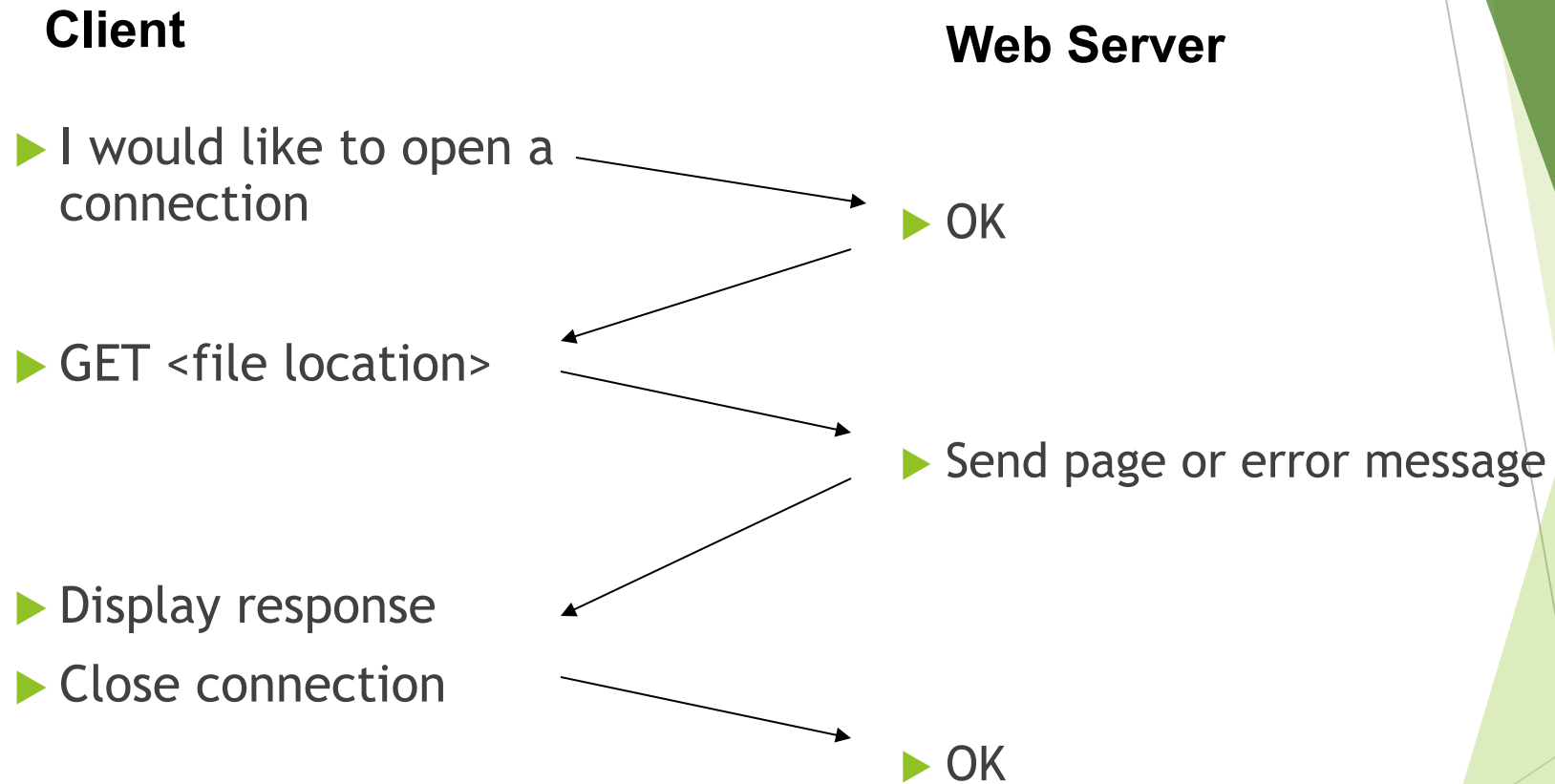


Summary

CEN 223 - Internet Communication

Assoc. Prof. Dr. Fatih ABUT
Department of Computer Engineering
Çukurova University

An HTTP conversation

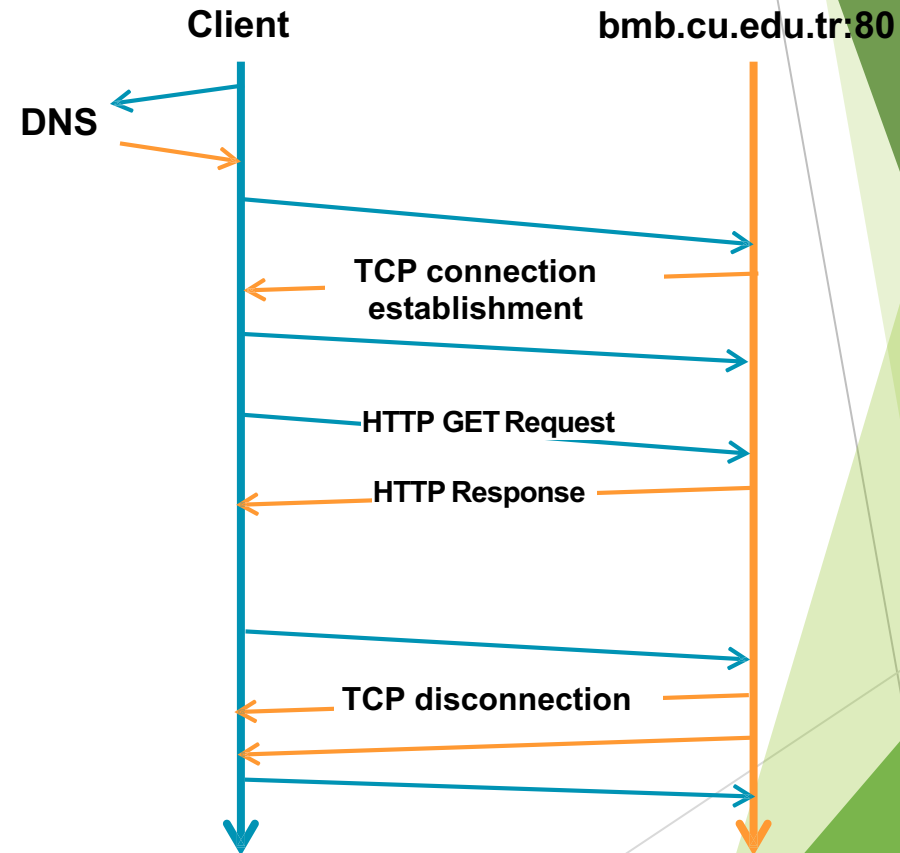


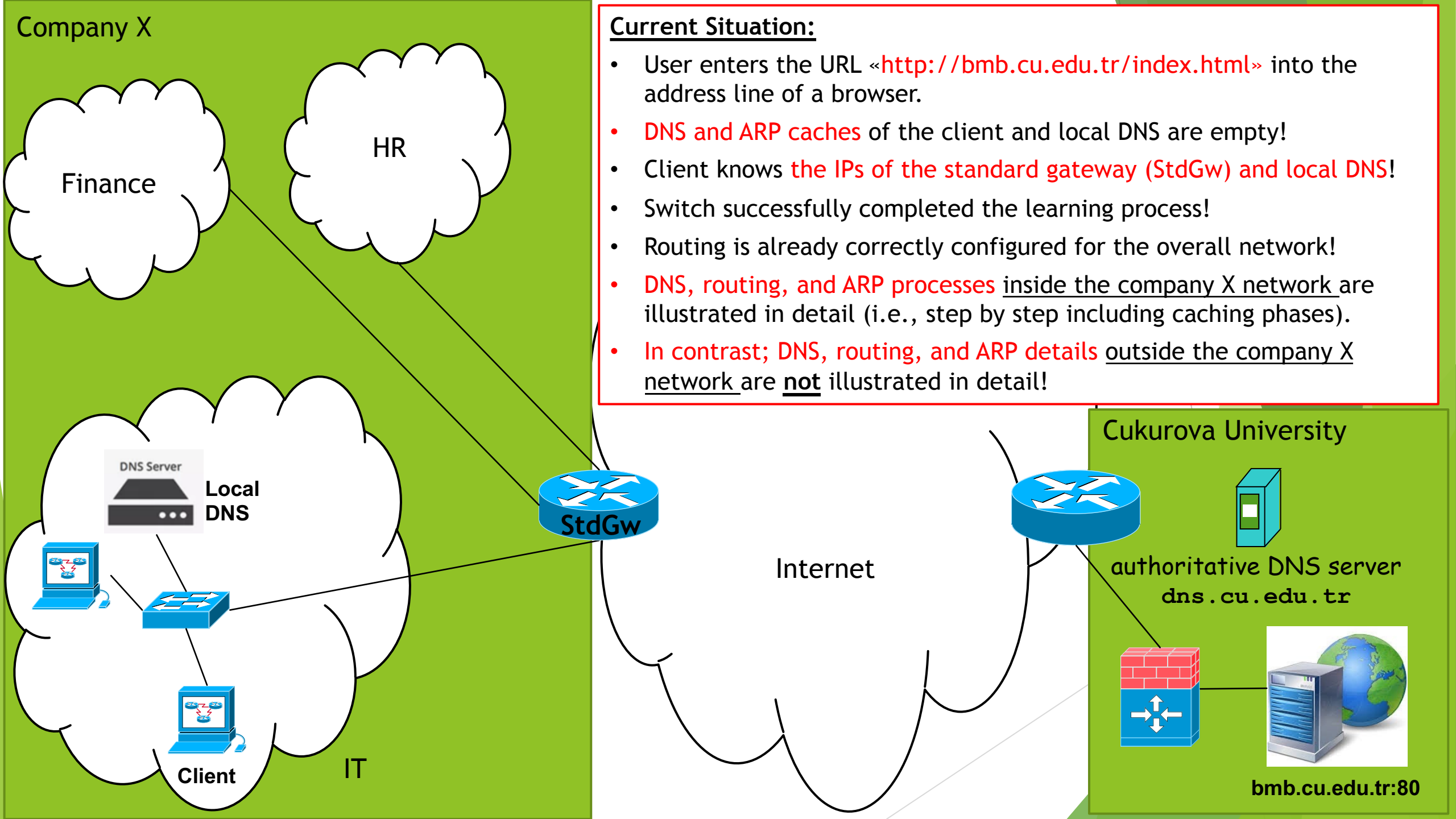
HTTP is the set of rules governing the format and content of the conversation between a Web client and server

Basic sequence of an HTTP request response

Example: Access to <http://bmb.cu.edu.tr:80/>

1. DNS resolution
2. Establishing a TCP connection to 193.140.54.17:80
3. Send HTTP GET request messages as bytestrome to 193.140.54.17:80.
4. Send HTTP response messages as bytestromes to the client.
5. Termination of the TCP connection by the client or server
6. Termination of the TCP connection by the server or client

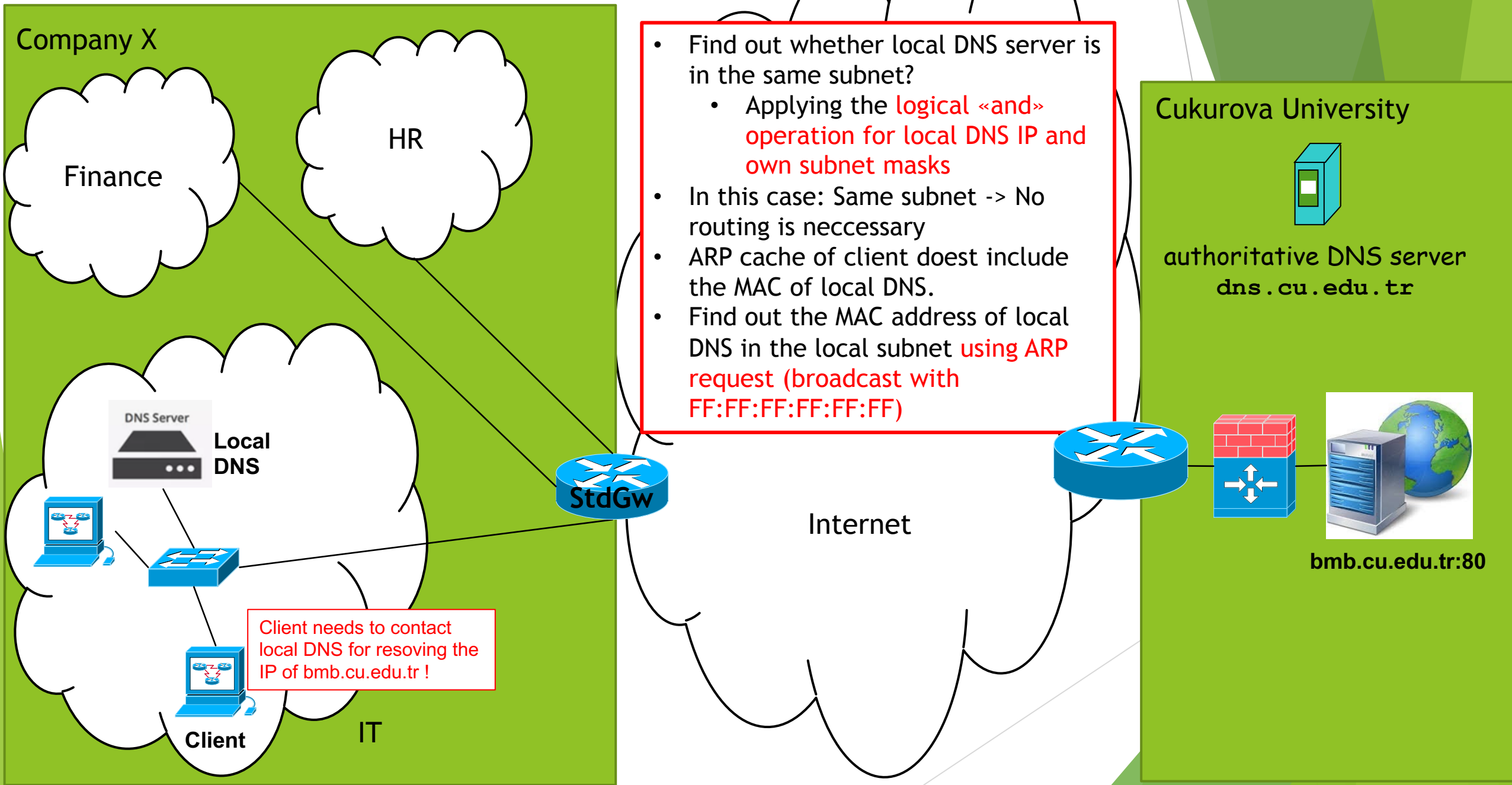




Current Situation:

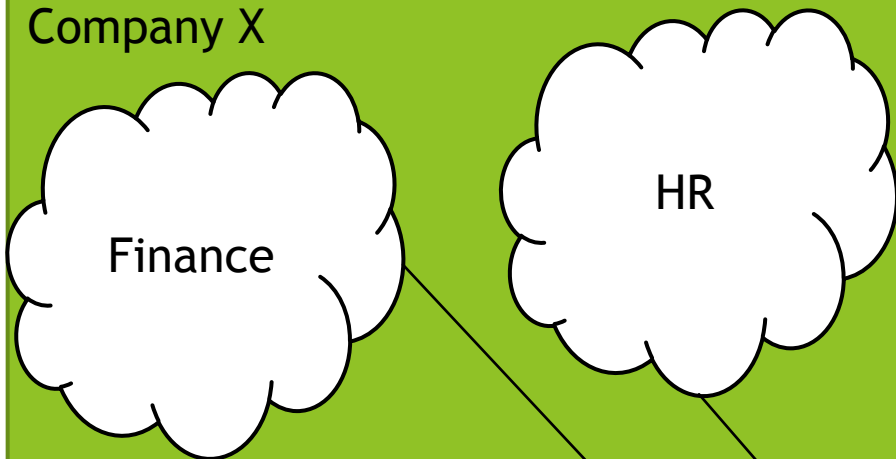
- User enters the URL <<http://bmb.cu.edu.tr/index.html>> into the address line of a browser.
- **DNS and ARP caches** of the client and local DNS are empty!
- Client knows **the IPs of the standard gateway (StdGw) and local DNS!**
- Switch successfully completed the learning process!
- Routing is already correctly configured for the overall network!
- **DNS, routing, and ARP processes** inside the company X network are illustrated in detail (i.e., step by step including caching phases).
- **In contrast; DNS, routing, and ARP details** outside the company X network are not illustrated in detail!

Step 1: Client wants to communicate with local DNS for resolving the IP of bmb.cu.edu.tr!
Check whether local DNS is in the same or different subnet!

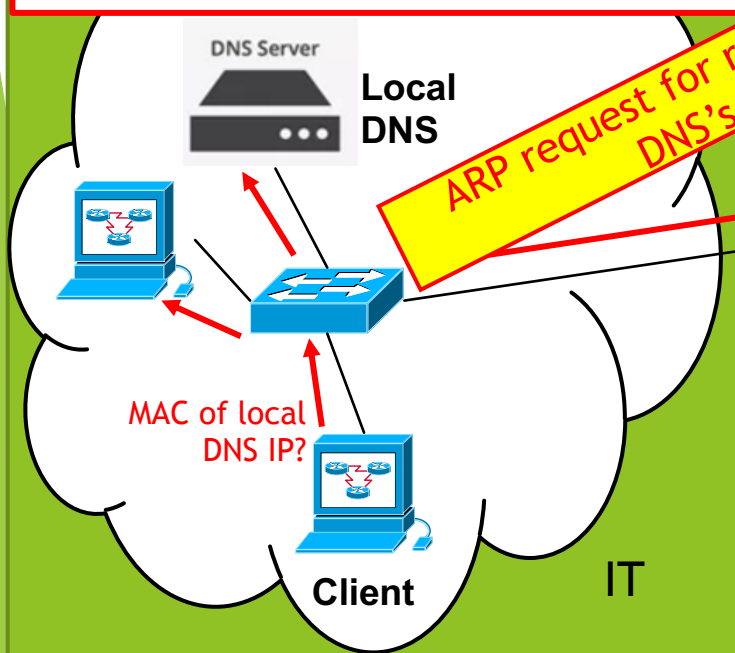


Step 2: Client resolves the MAC of local DNS IP (Request)

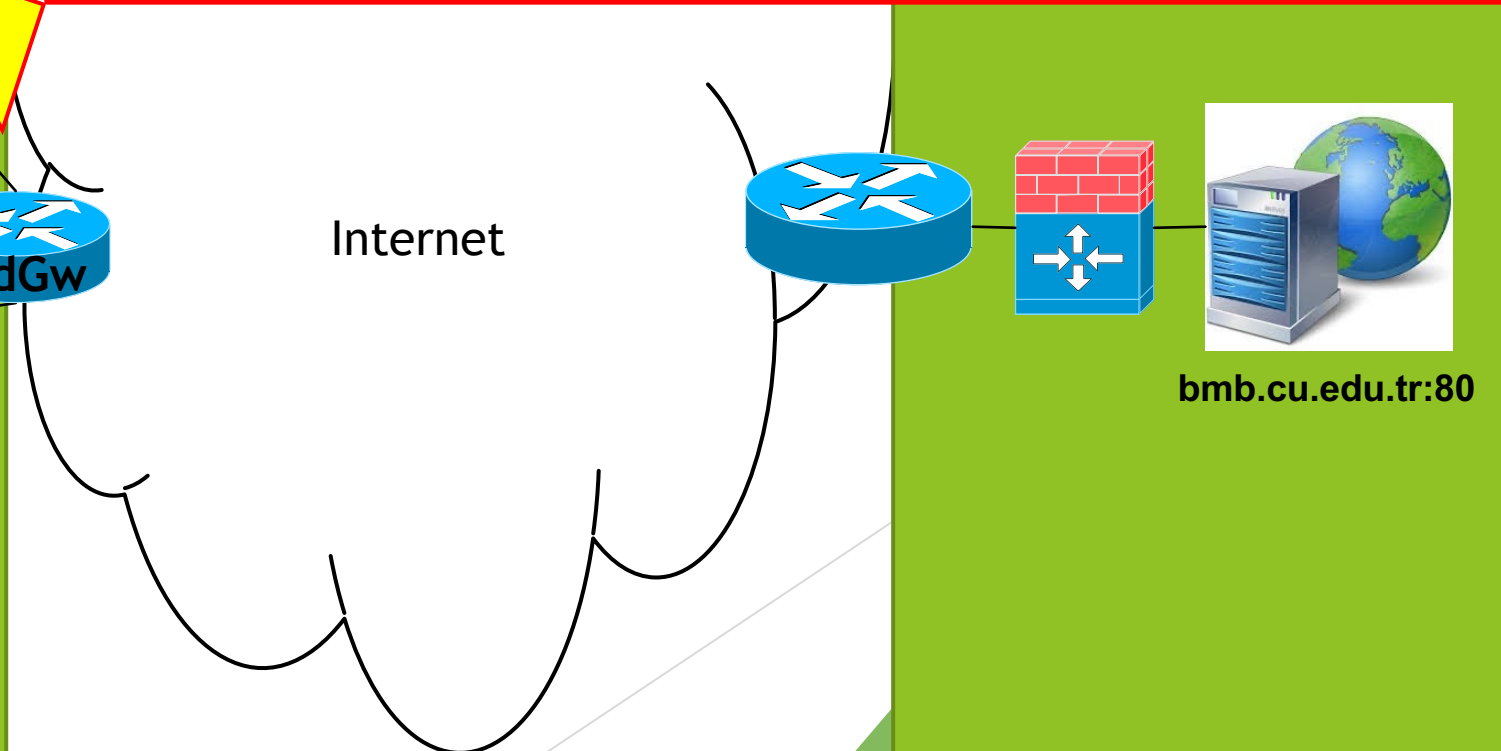
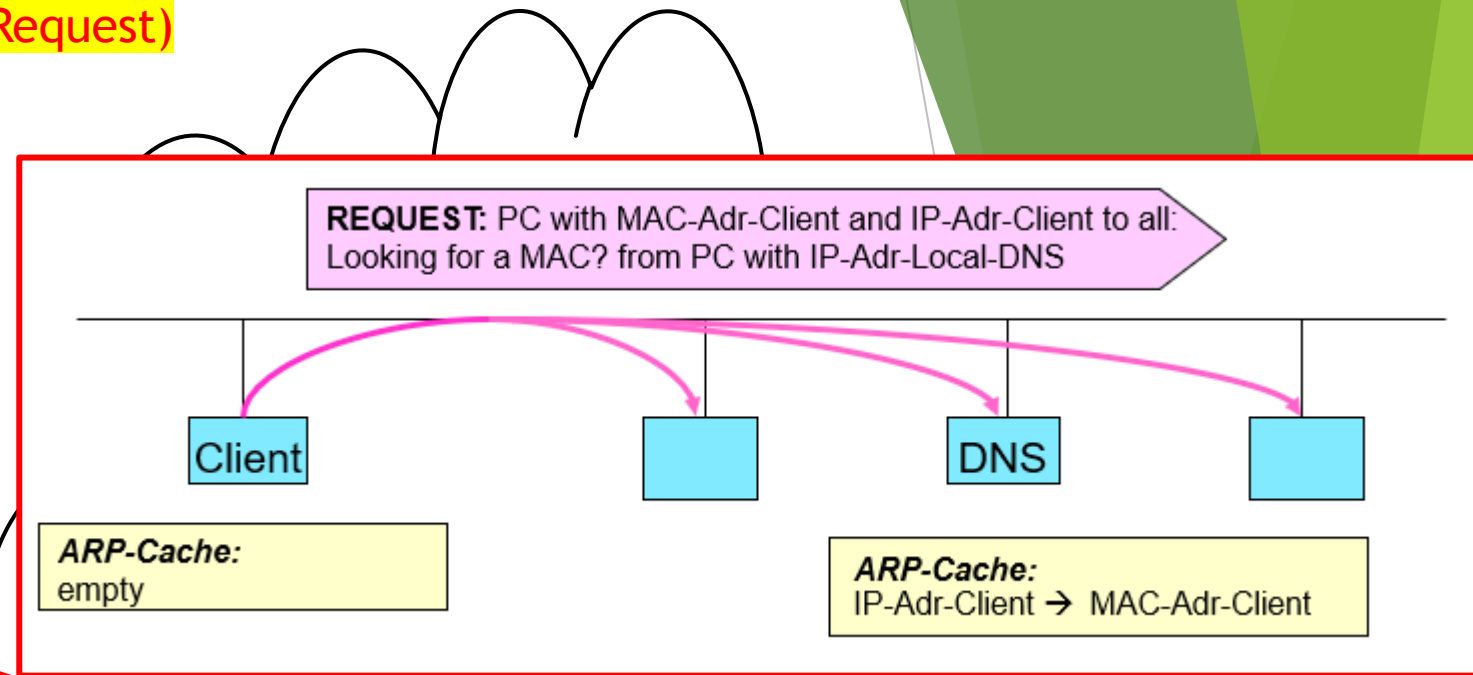
Company X



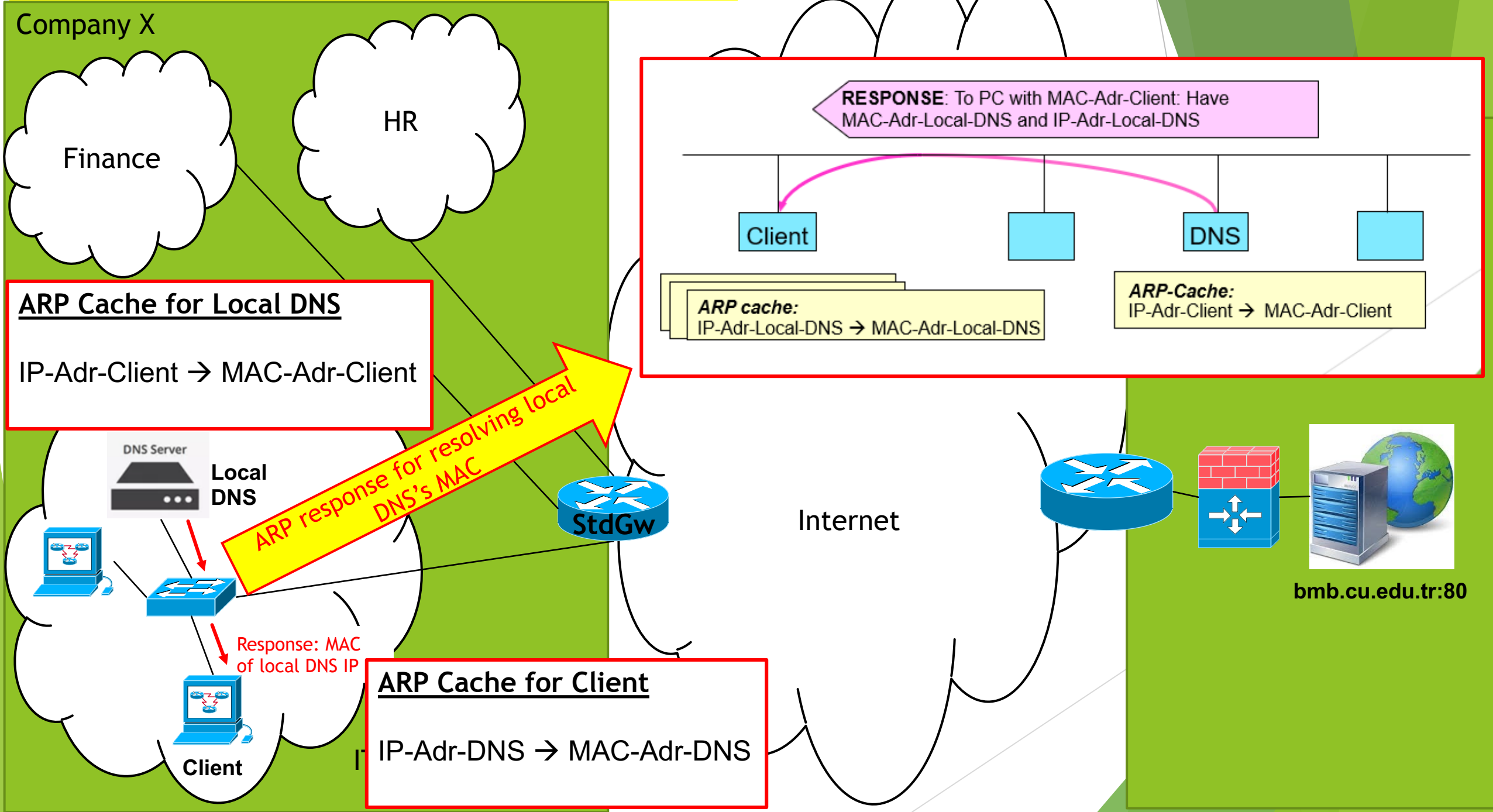
ARP Cache for Local DNS
IP-Adr-Client → MAC-Adr-Client



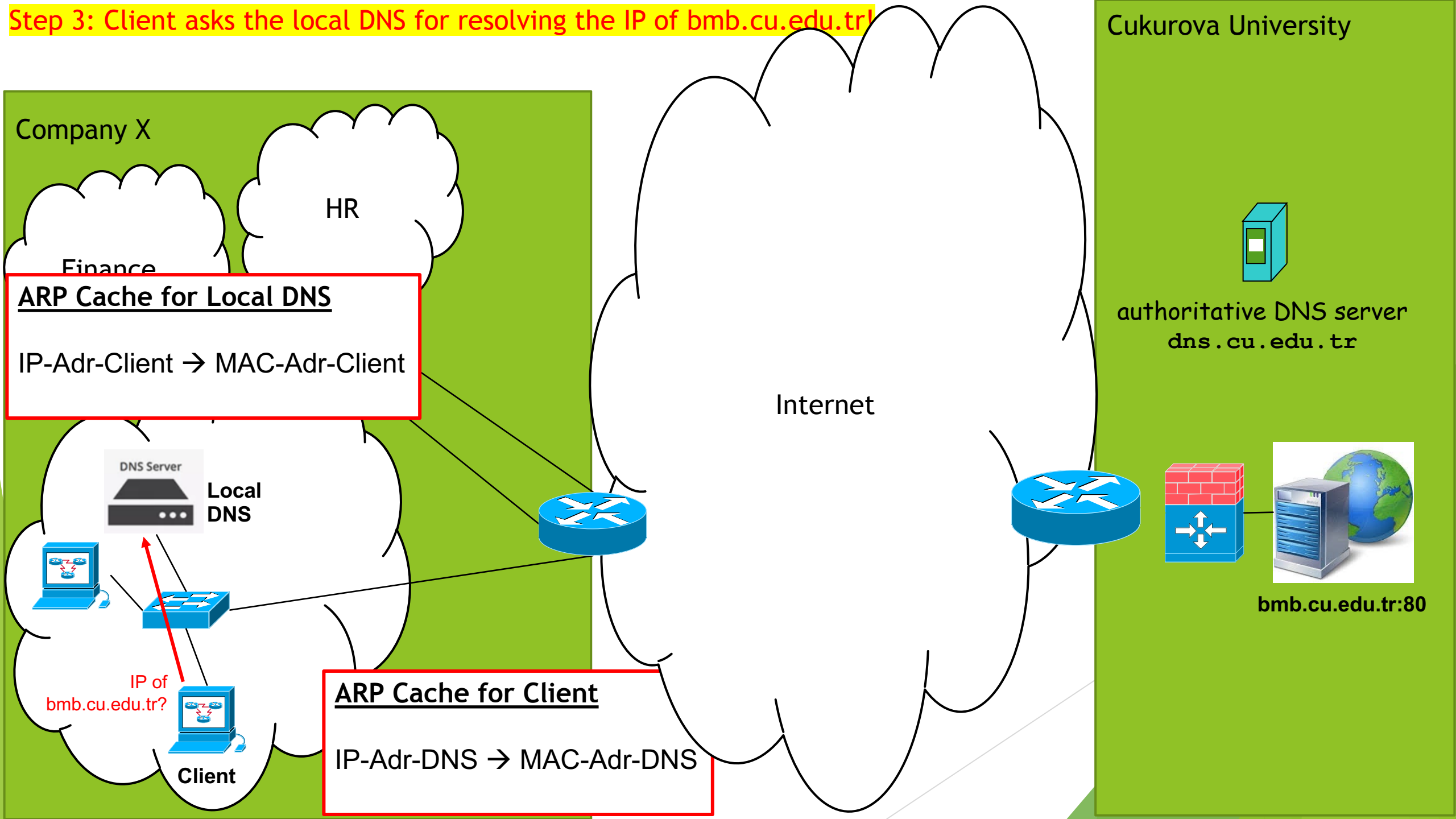
ARP request for resolving local DNS's MAC



Step 2: Client resolves the MAC of local DNS IP (Response)

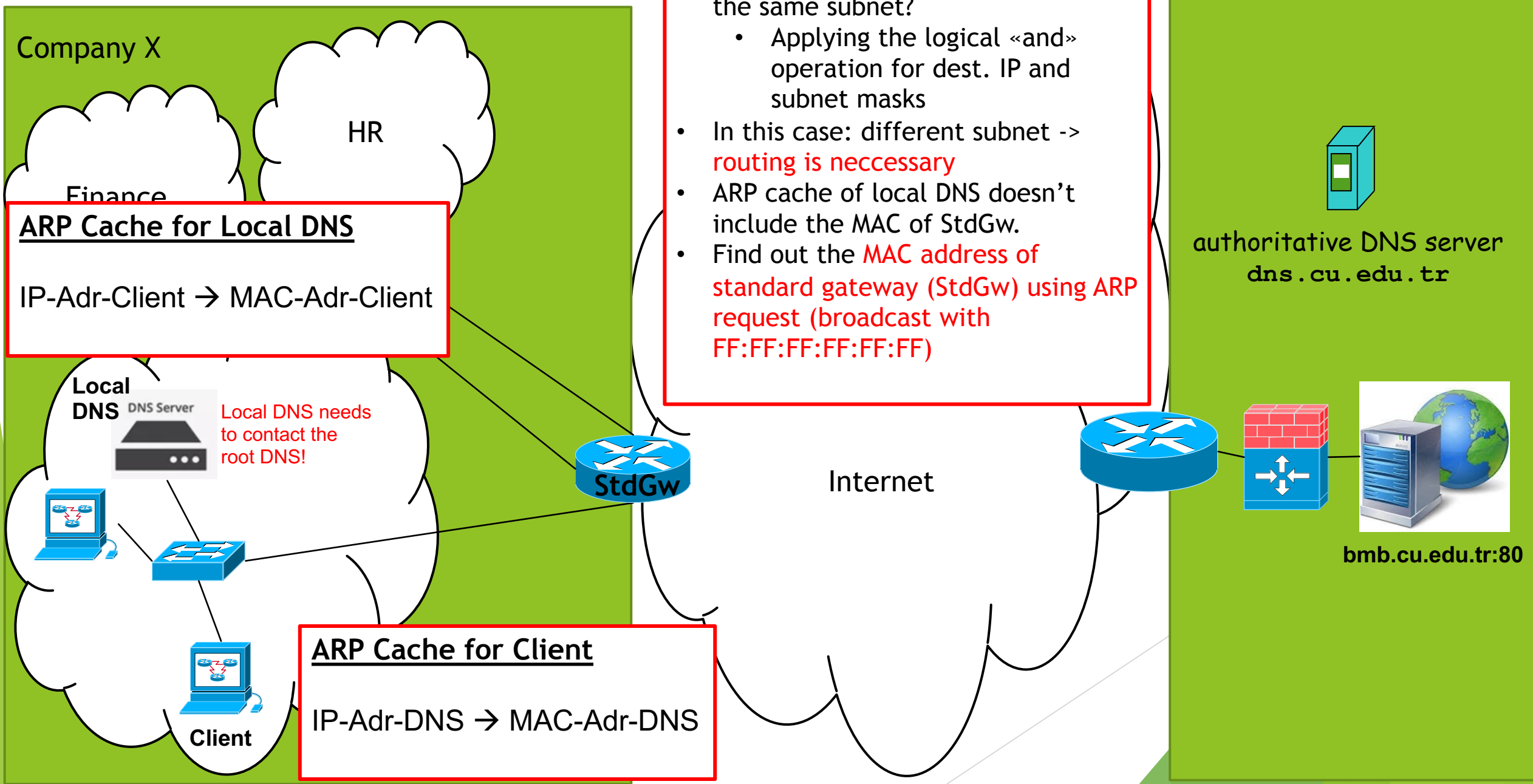


Step 3: Client asks the local DNS for resolving the IP of bmb.cu.edu.tr!

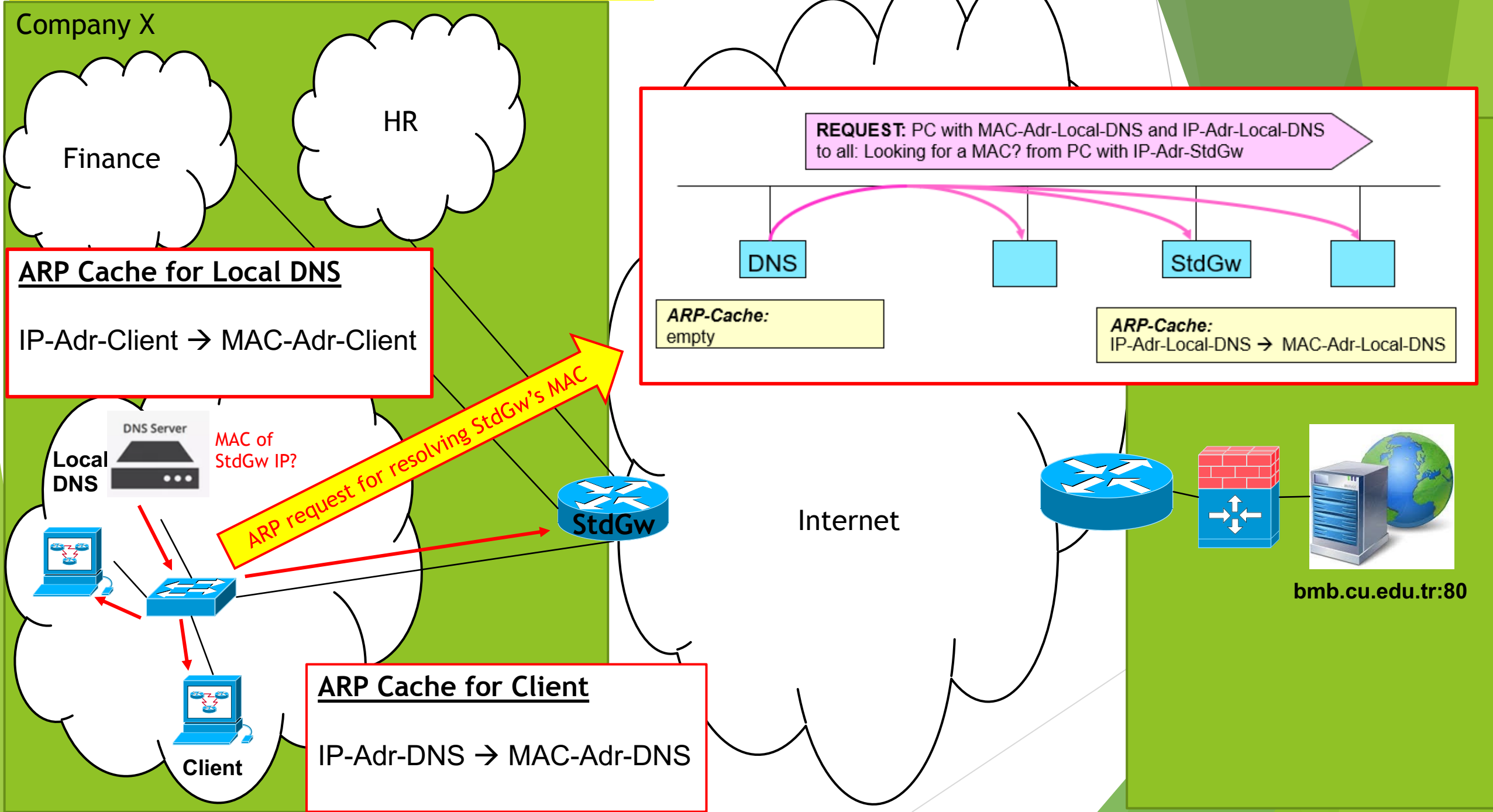


Step 4: Local DNS needs to communicate with root DNS for resolving the IP of bmb.cu.edu.tr!

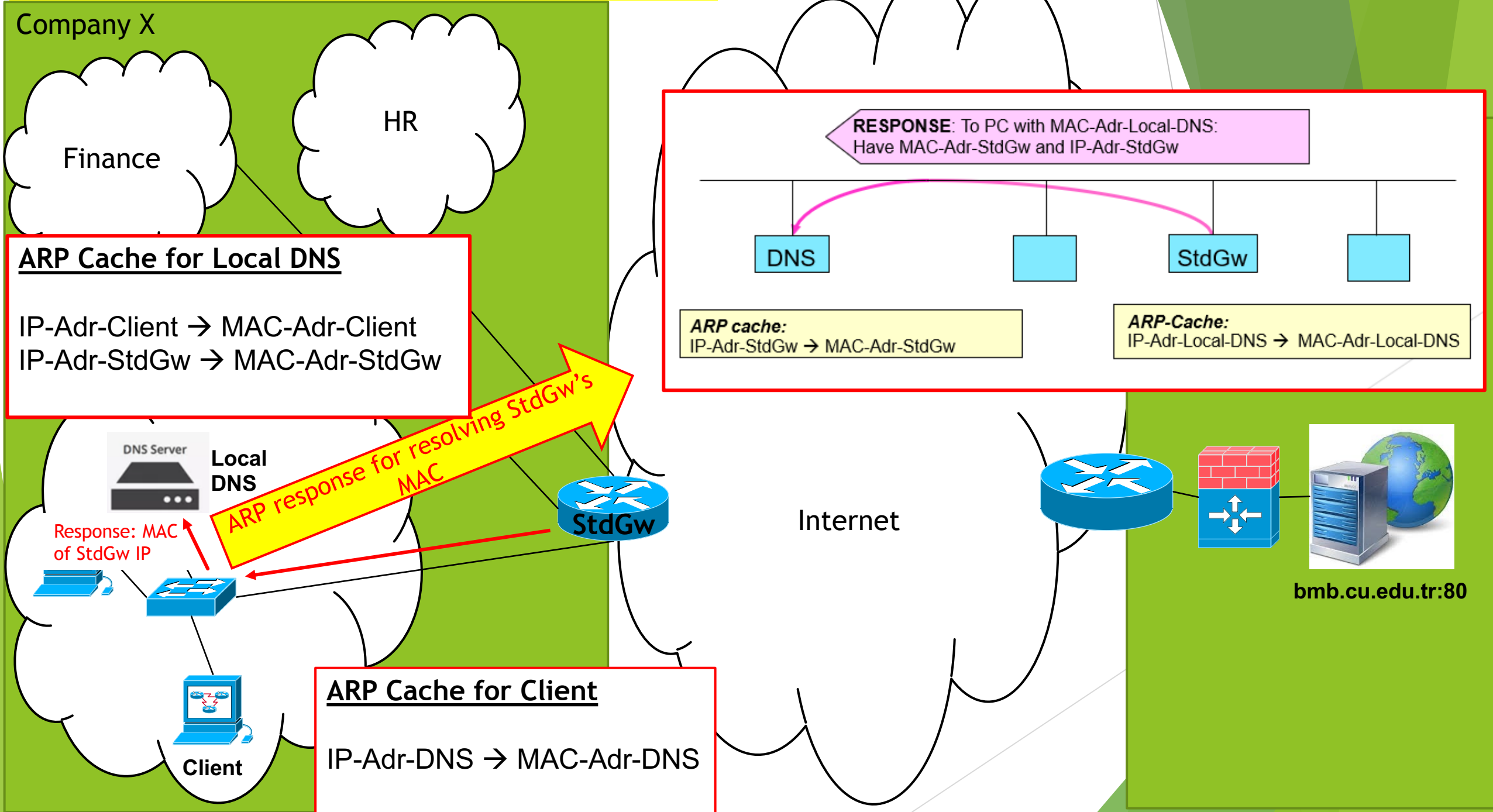
Check whether root DNS is in the same or different subnet!



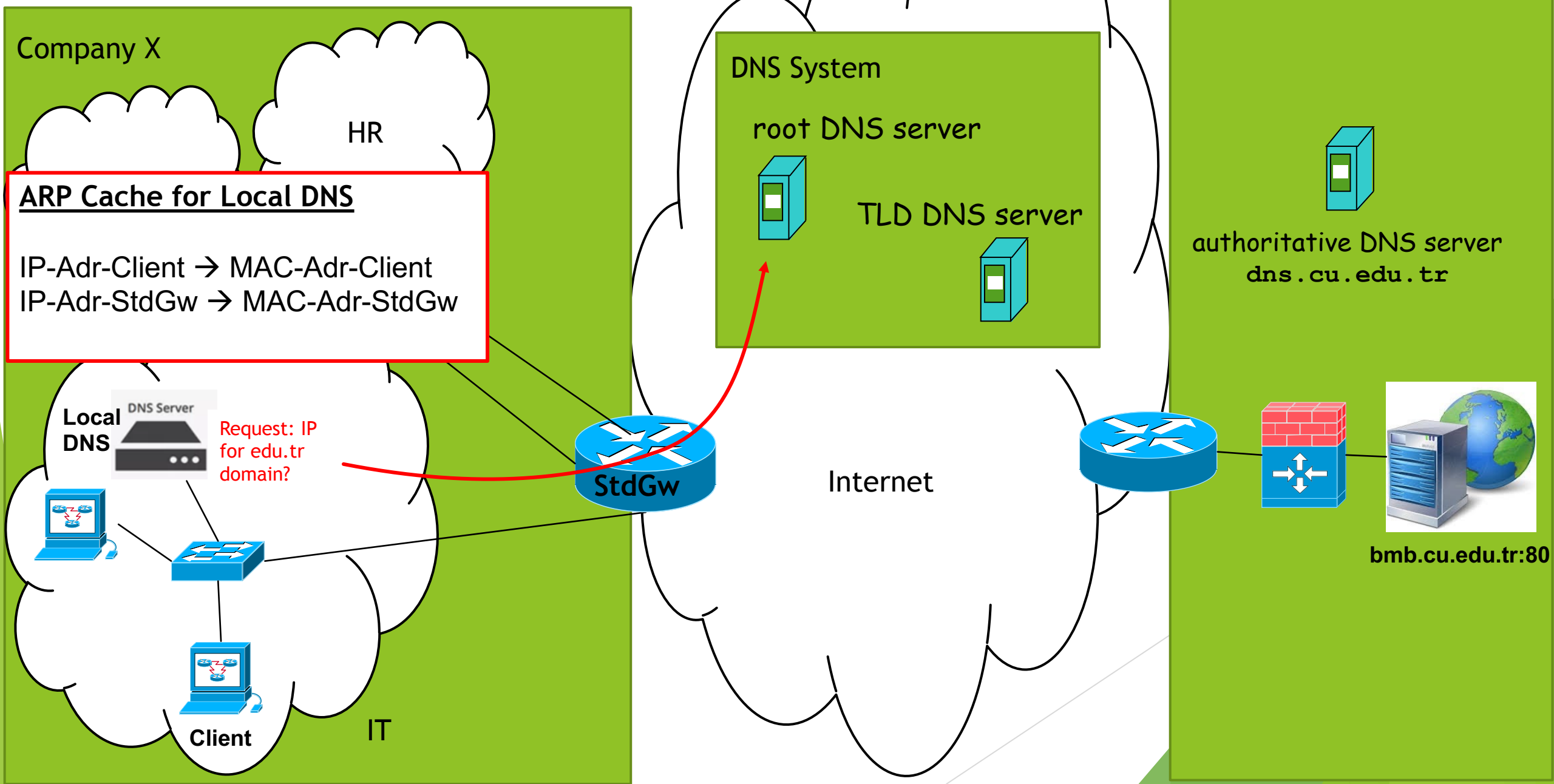
Step 4: Client resolves the MAC of StdGw (Request)



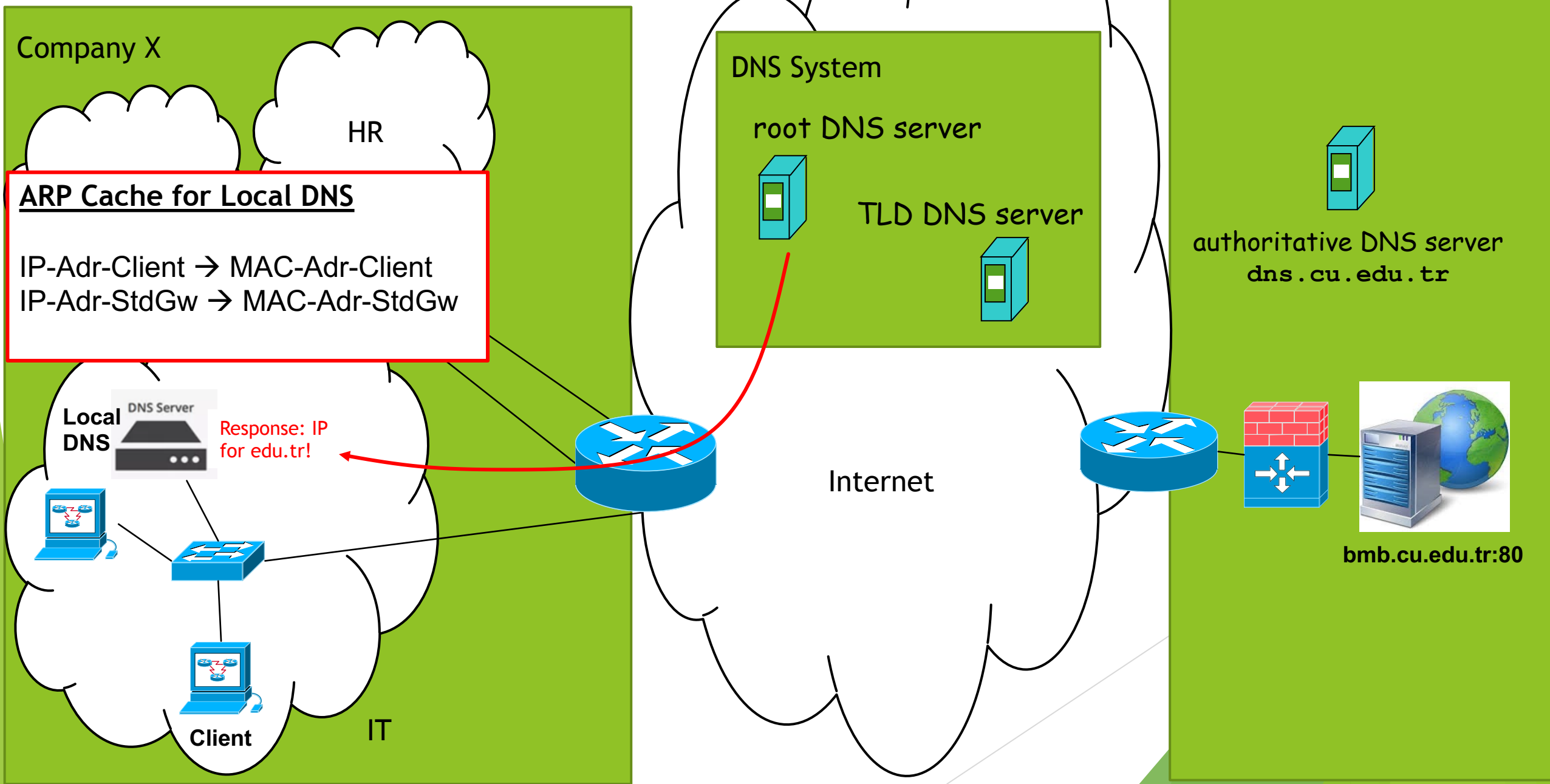
Step 4: Local DNS resolves the MAC of StdGw (Response)



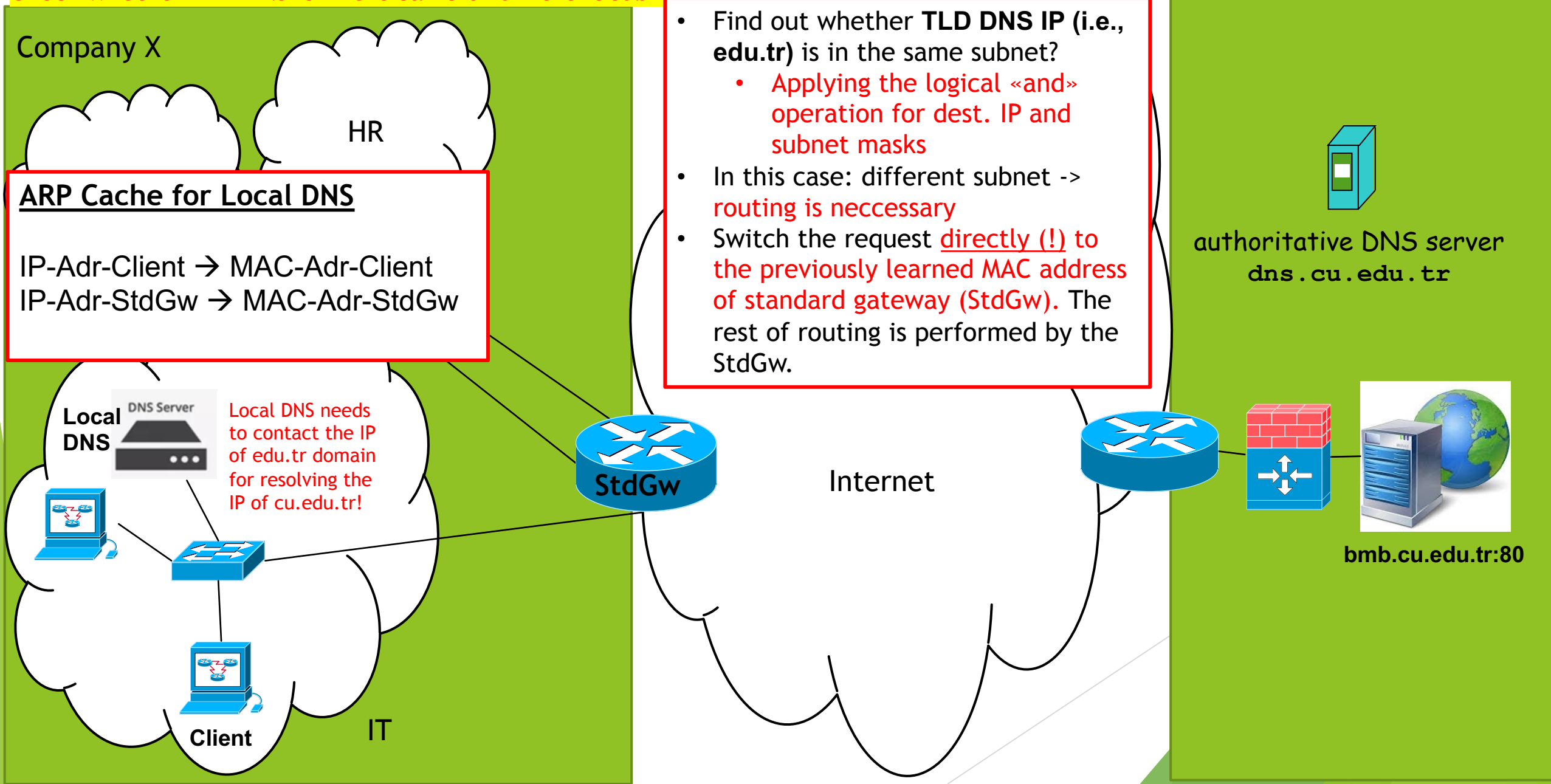
Step 5: Local DNS asks the root DNS for the IP of edu.tr over the StdGw! (StdGw takes the rest of routing!)



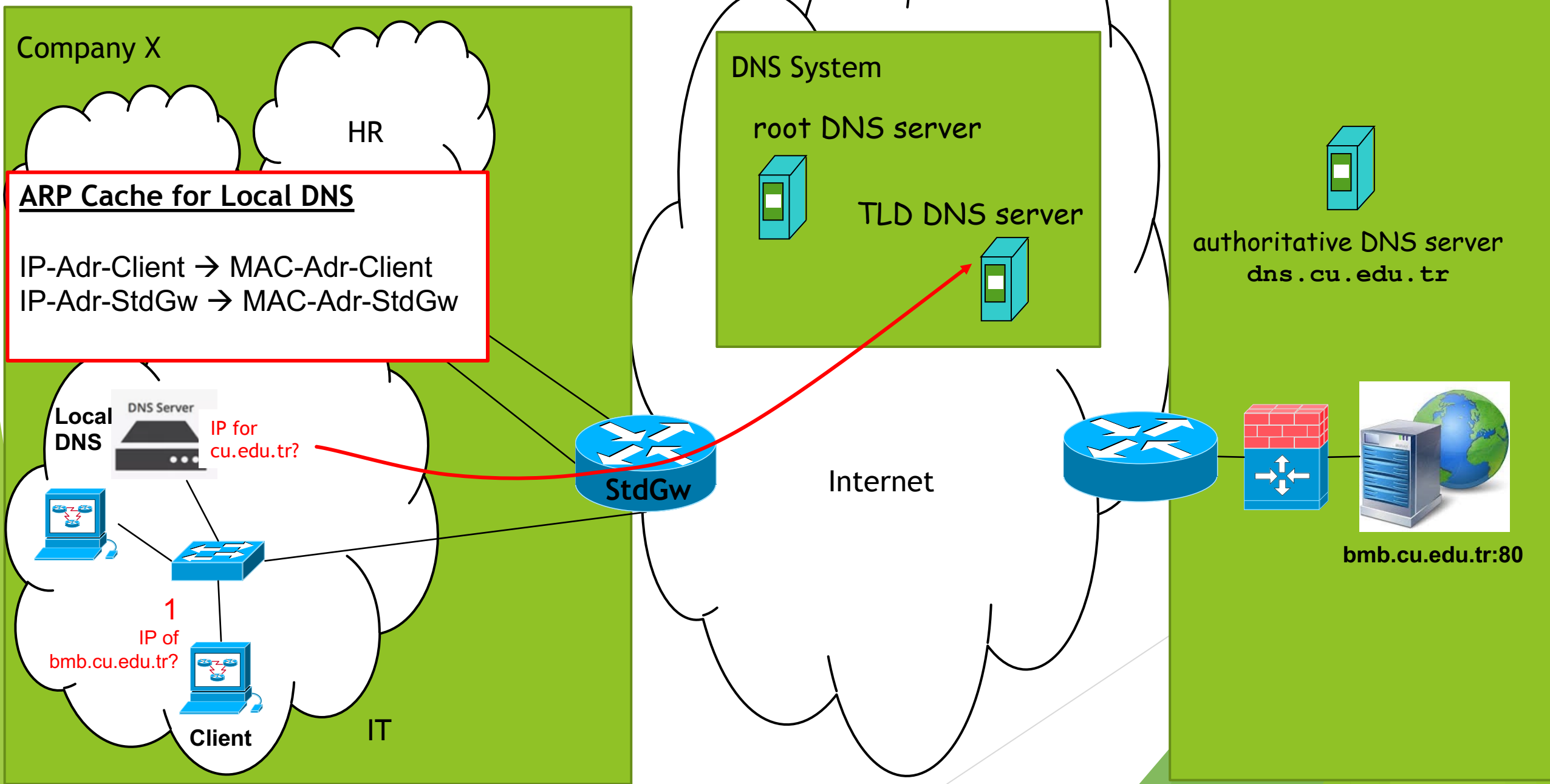
Step 5: Local DNS receives the IP of edu.tr from the root DNS !



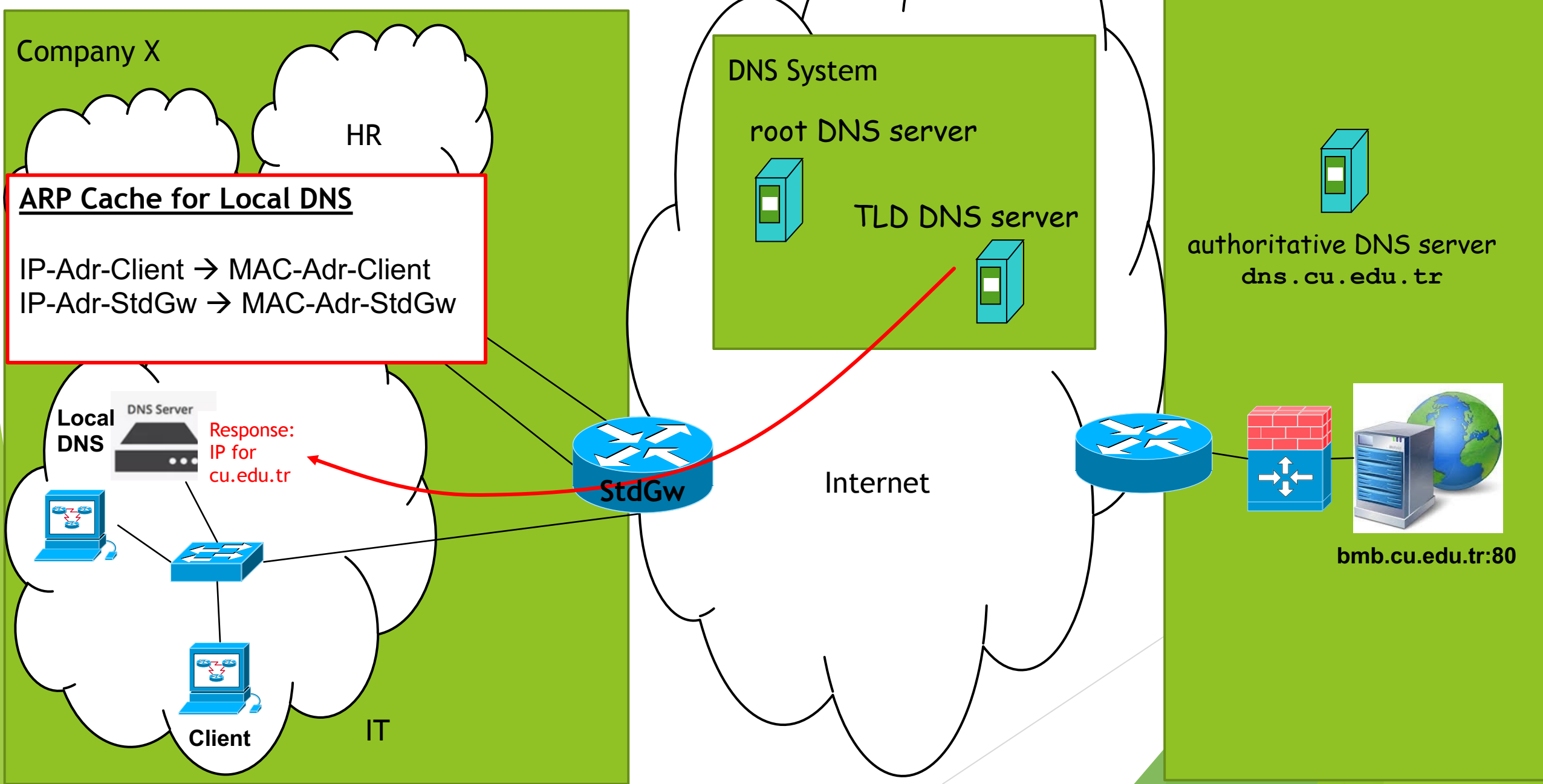
Step 6: Local DNS needs to communicate with TLD DNS for resolving the IP of cu.edu.tr domain.
Check whether TLD DNS is in the same or different subnet!



Step 6: Local DNS asks the TLD DNS for the IP of cu.edu.tr over the StdGw.



Step 7: Local DNS receives from the TLD DNS the IP of cu.edu.tr!



Step 8: Local DNS needs to communicate with Authoritative DNS for resolving the IP of **bmb.cu.edu.tr**. Check whether Authoritative DNS is in the same or different subnet!

Company X

HR

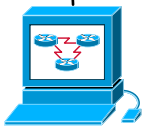
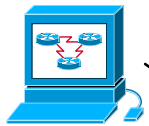
ARP Cache for Local DNS

IP-Adr-Client → MAC-Adr-Client
IP-Adr-StdGw → MAC-Adr-StdGw

Local
DNS



Local DNS needs to contact the IP of **cu.edu.tr** domain for resolving the IP of **bmb.cu.edu.tr**!



Client

IT

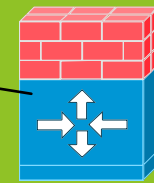
StdGw

Internet

- Find out whether **Authoritative DNS IP (cu.edu.tr)** is in the same subnet?
 - Applying the logical «and» operation for IP of authoritative DNS and own subnet masks
- In this case: different subnet -> **routing is necessary**
- Switch the request **directly (!)** to the **MAC address of standard gateway (StdGw)**. The rest of routing is performed by the StdGw.

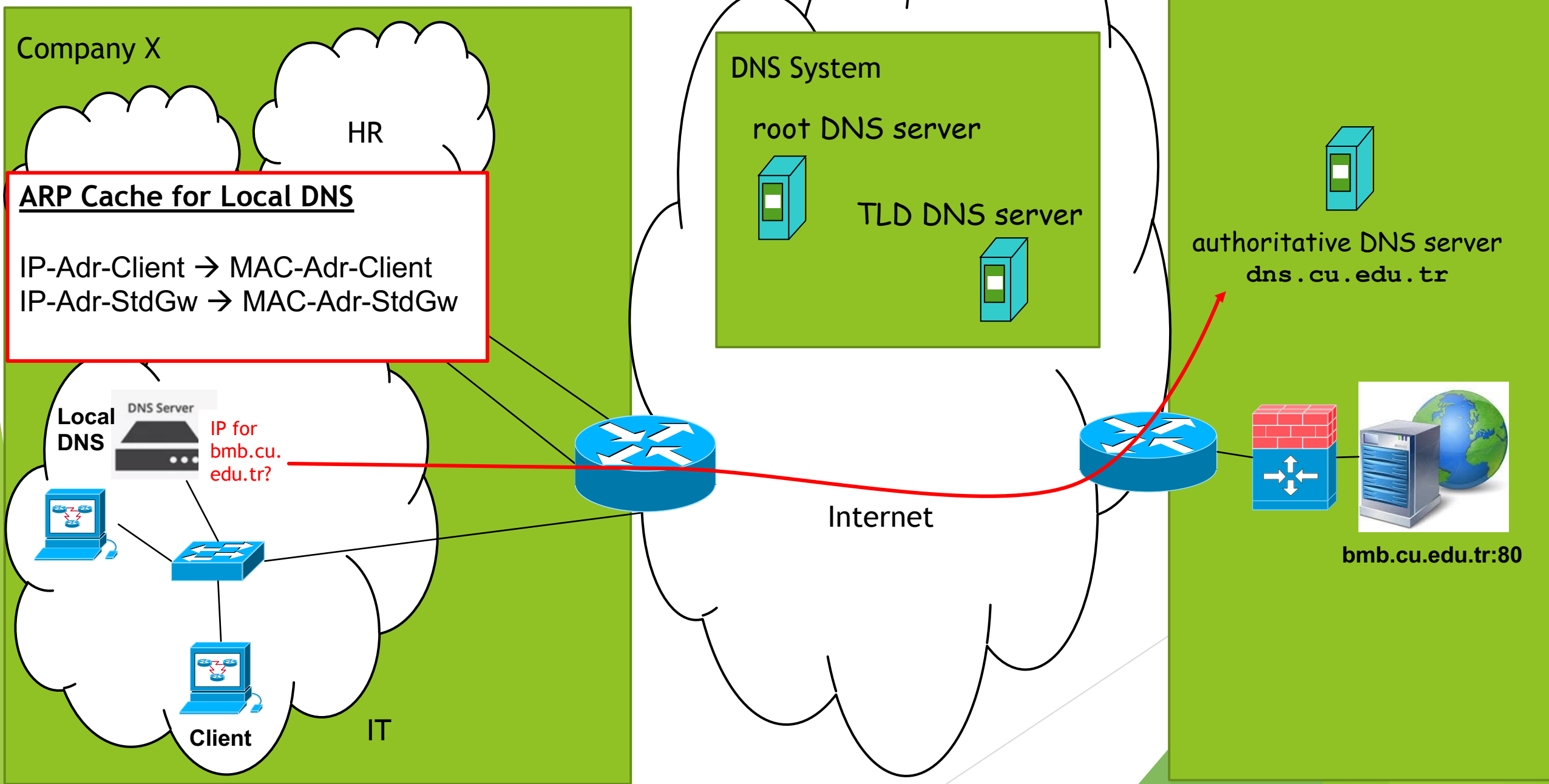


authoritative DNS server
dns.cu.edu.tr

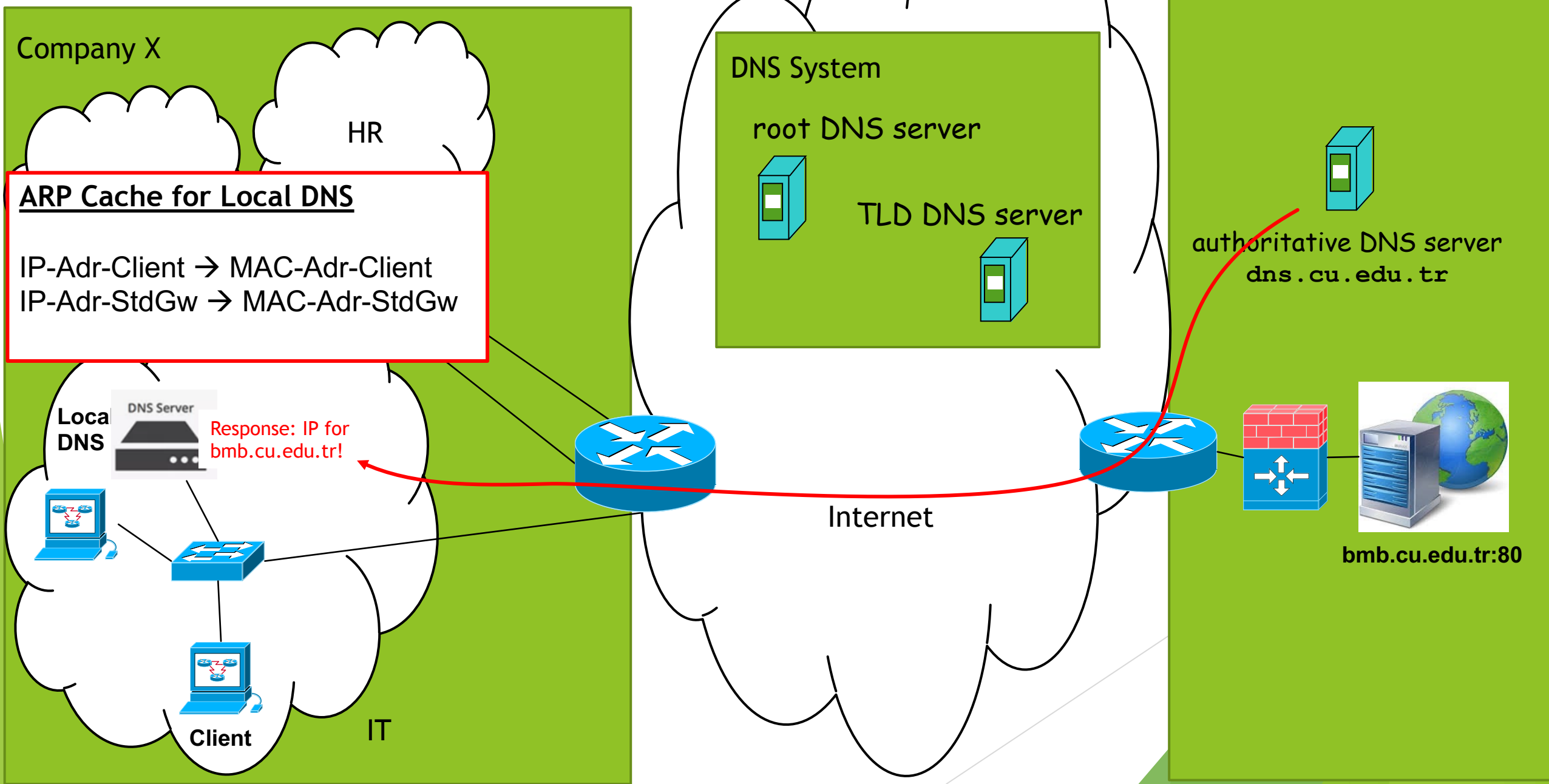


bmb.cu.edu.tr:80

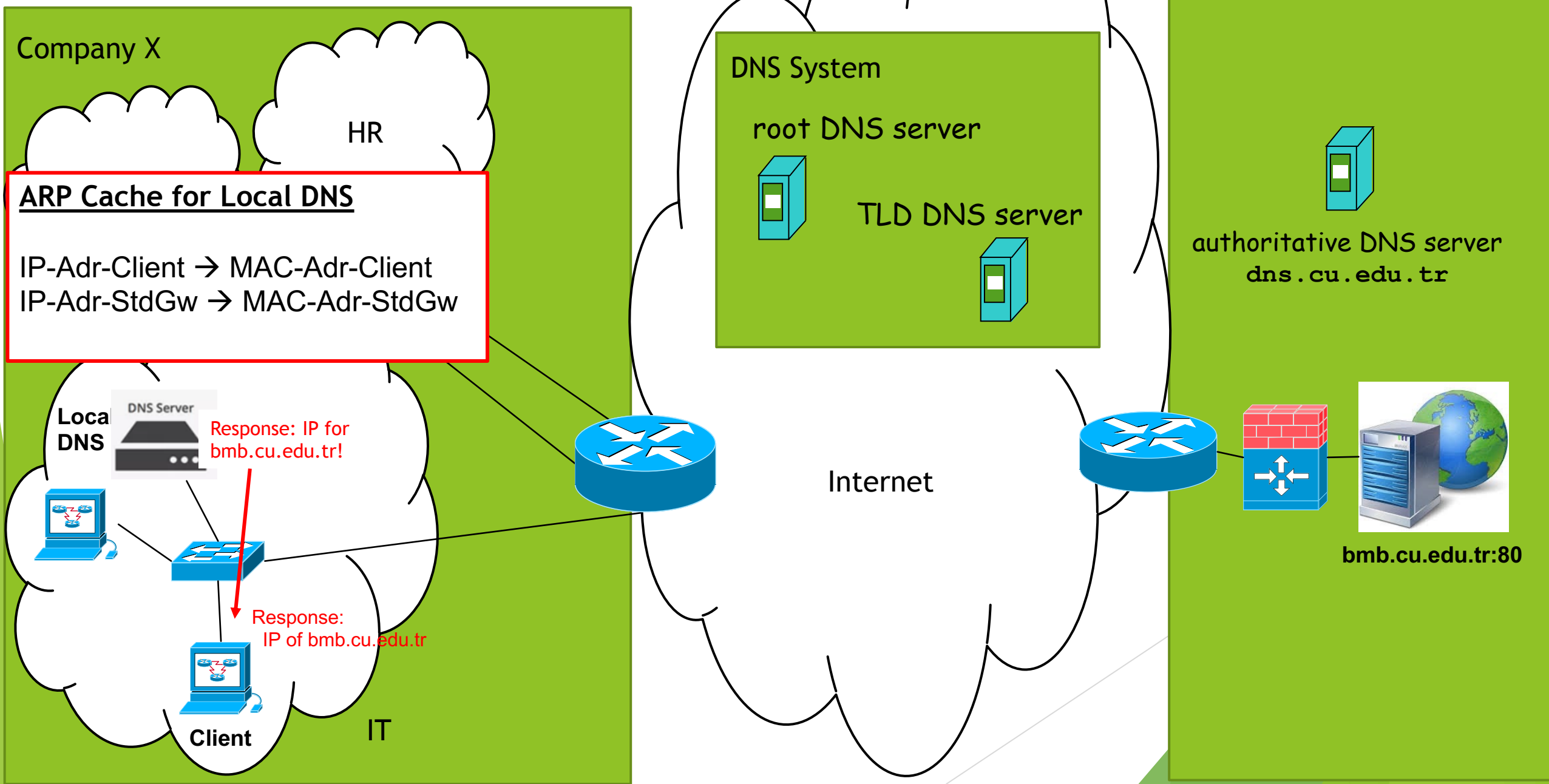
Step 8: Local DNS asks the Authoritative DNS for the IP of bmb.cu.edu.tr over the StdGw.



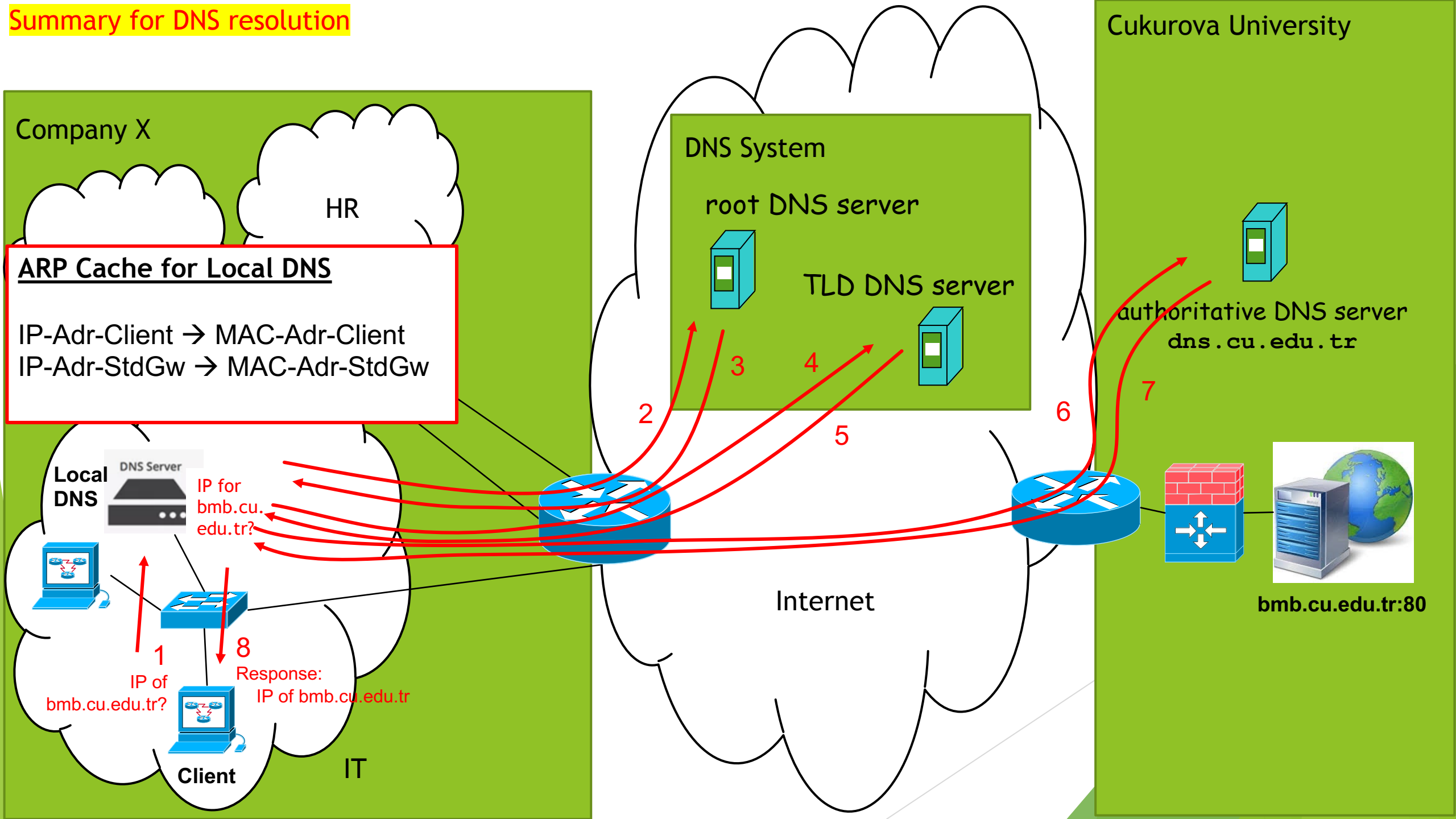
Step 9: Local DNS receives the IP of bmb.cu.edu.tr from the Authoritative DNS !



Step 10: Local DNS sends the IP of bmb.cu.edu.tr to the client.

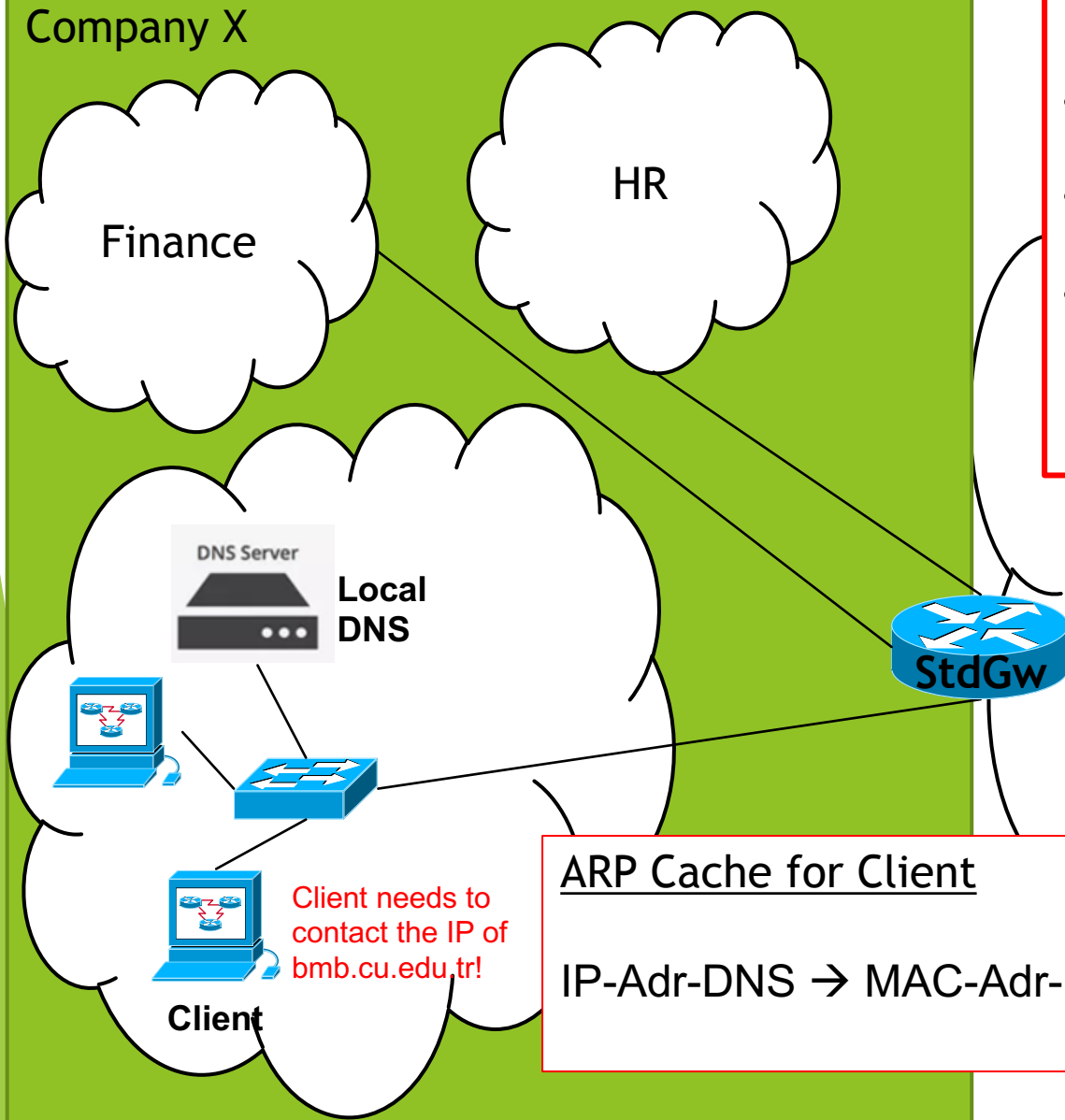


Summary for DNS resolution



Step 11: Client wants to establish a TCP connection to the IP of bmb.cu.edu.tr.
Check whether IP of bmb.cu.edu.tr is in the same or different subnet!

Company X



- Find out whether IP of **bmb.cu.edu.tr** is in the same subnet?
 - Applying the logical «and» operation for dest. IP and subnet masks
- In this case: different subnet -> **routing is necessary**
- ARP cache of client doesn't include the MAC of StdGw.
- Find out the **MAC address of standard gateway (StdGw) using ARP request (broadcast FF:FF:FF:FF:FF:FF)**

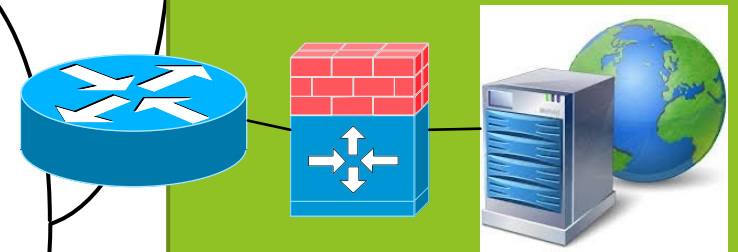
ARP Cache for Client

IP-Adr-DNS → MAC-Adr-DNS

Cukurova University

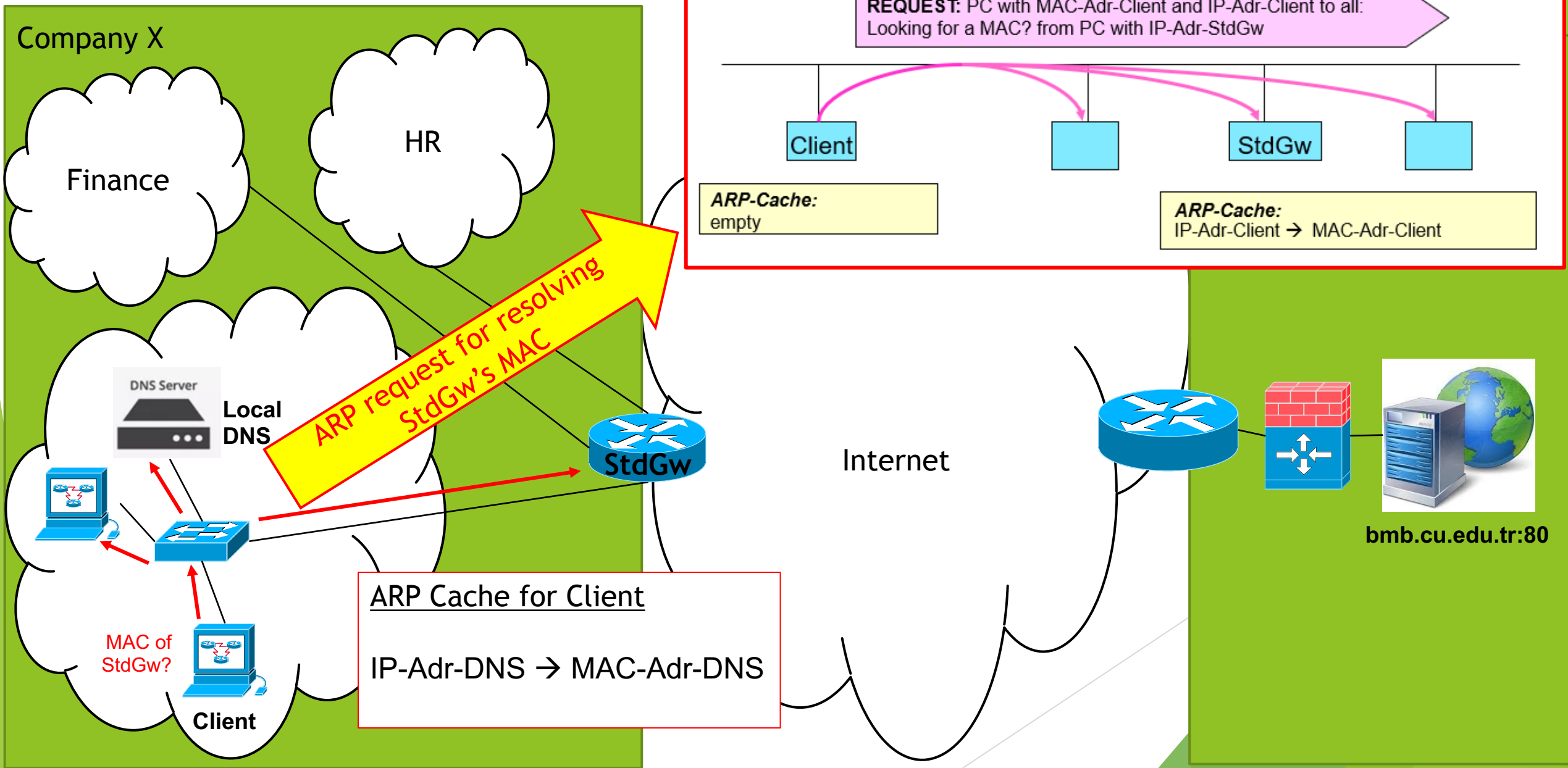


authoritative DNS server
`dns.cu.edu.tr`

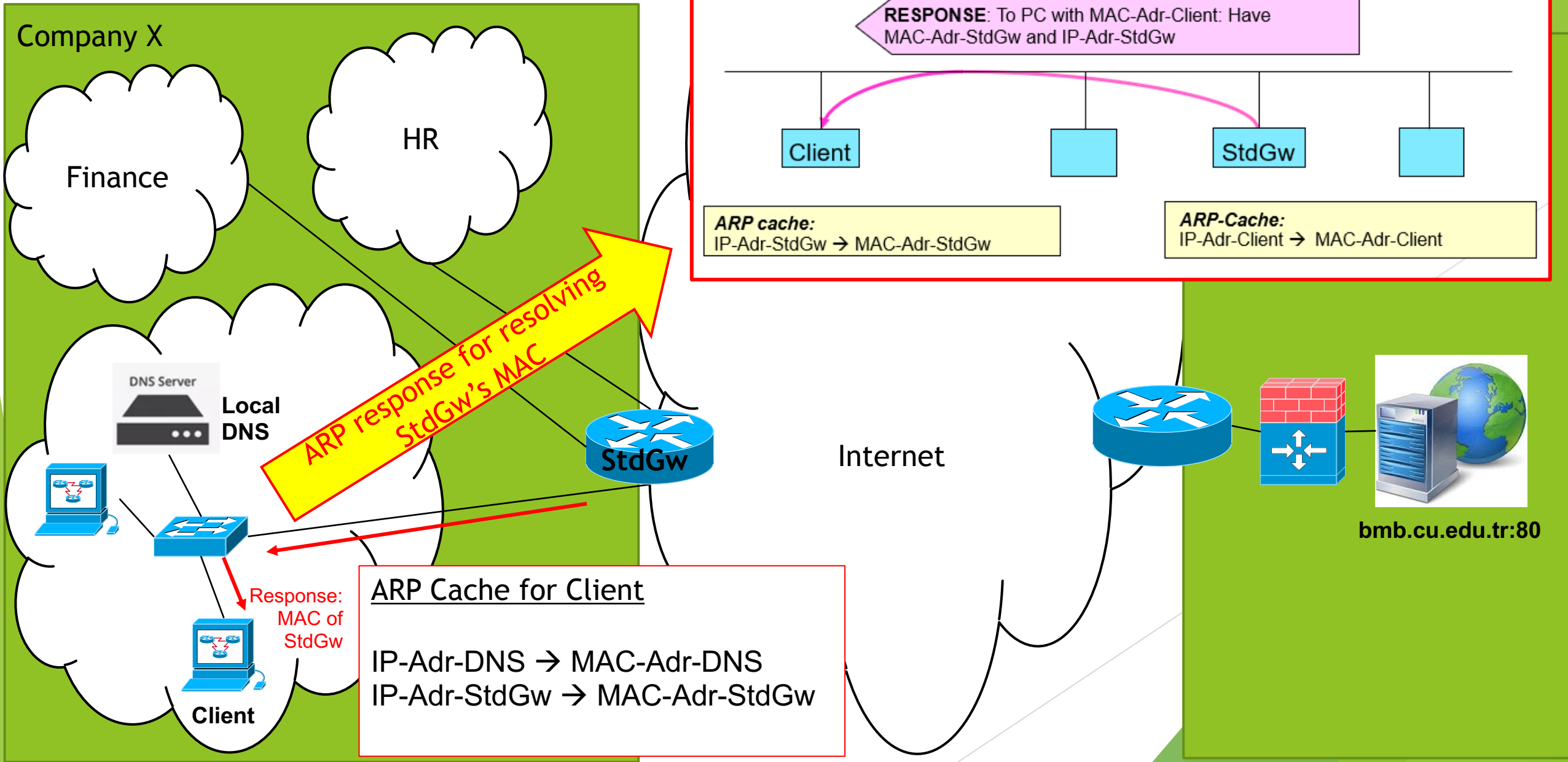


bmb.cu.edu.tr:80

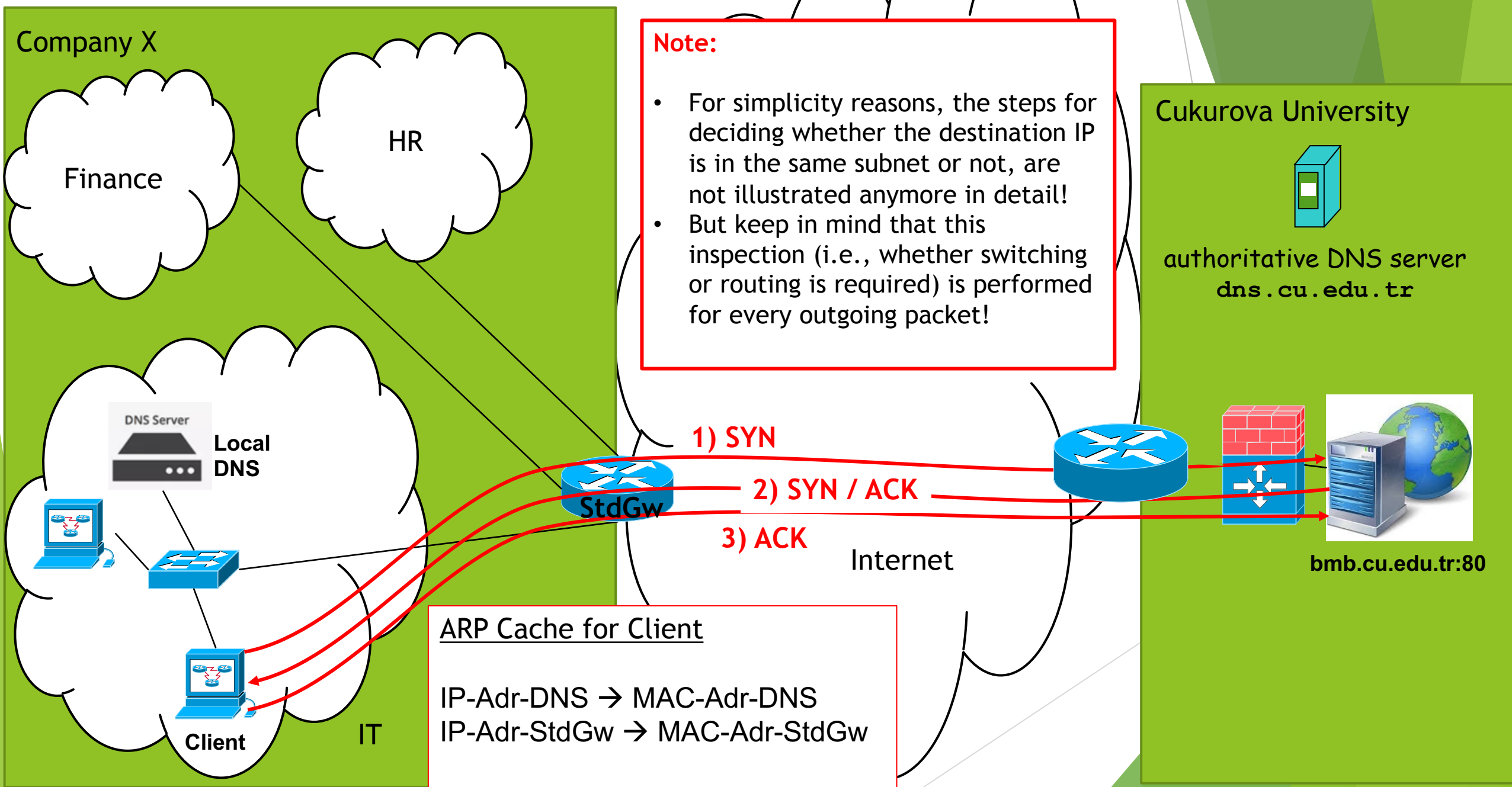
Step 11: Client resolves the MAC of StdGw IP (Request)



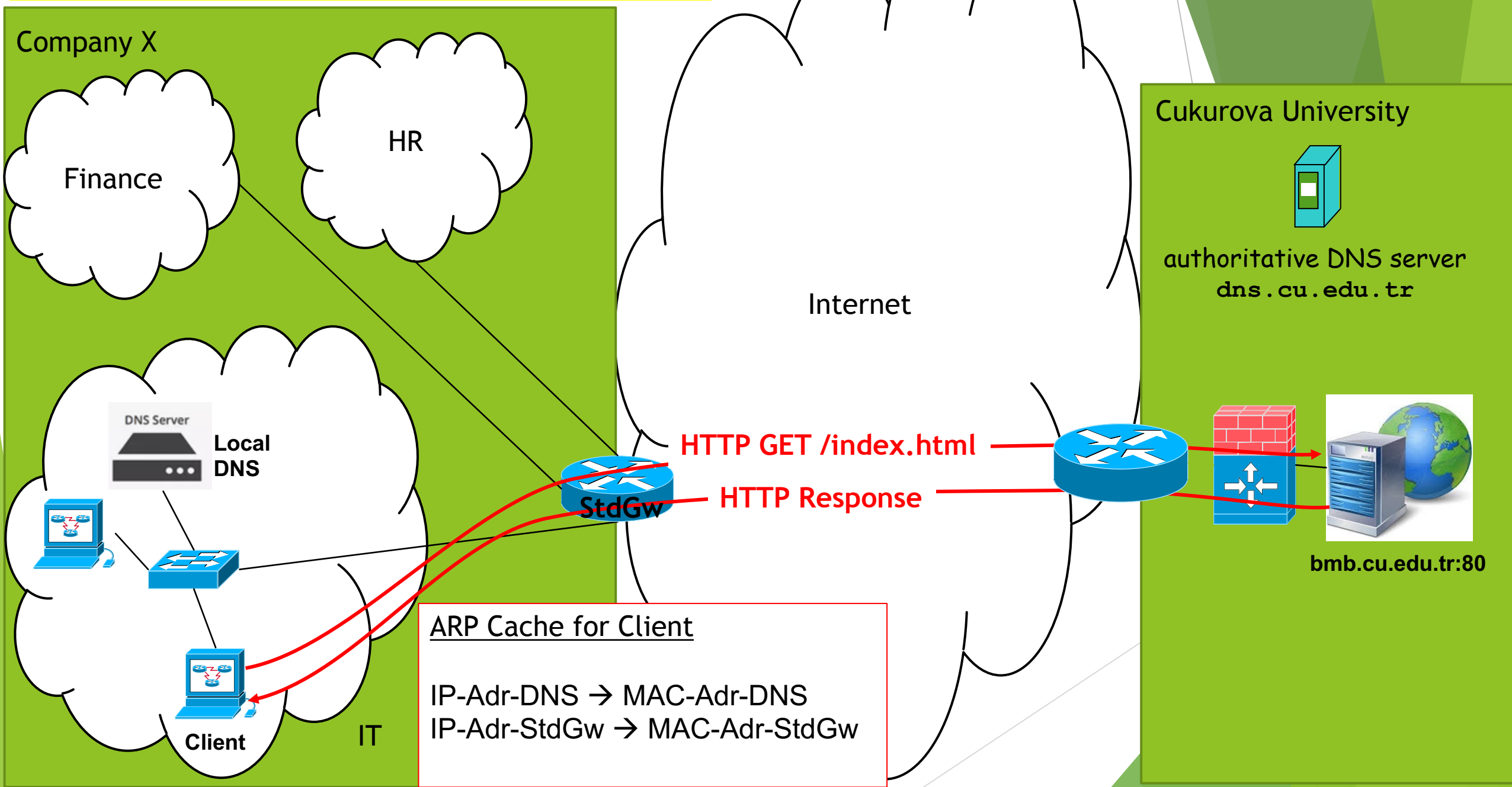
Step 12: Client resolves the MAC of StdGw IP (Response)



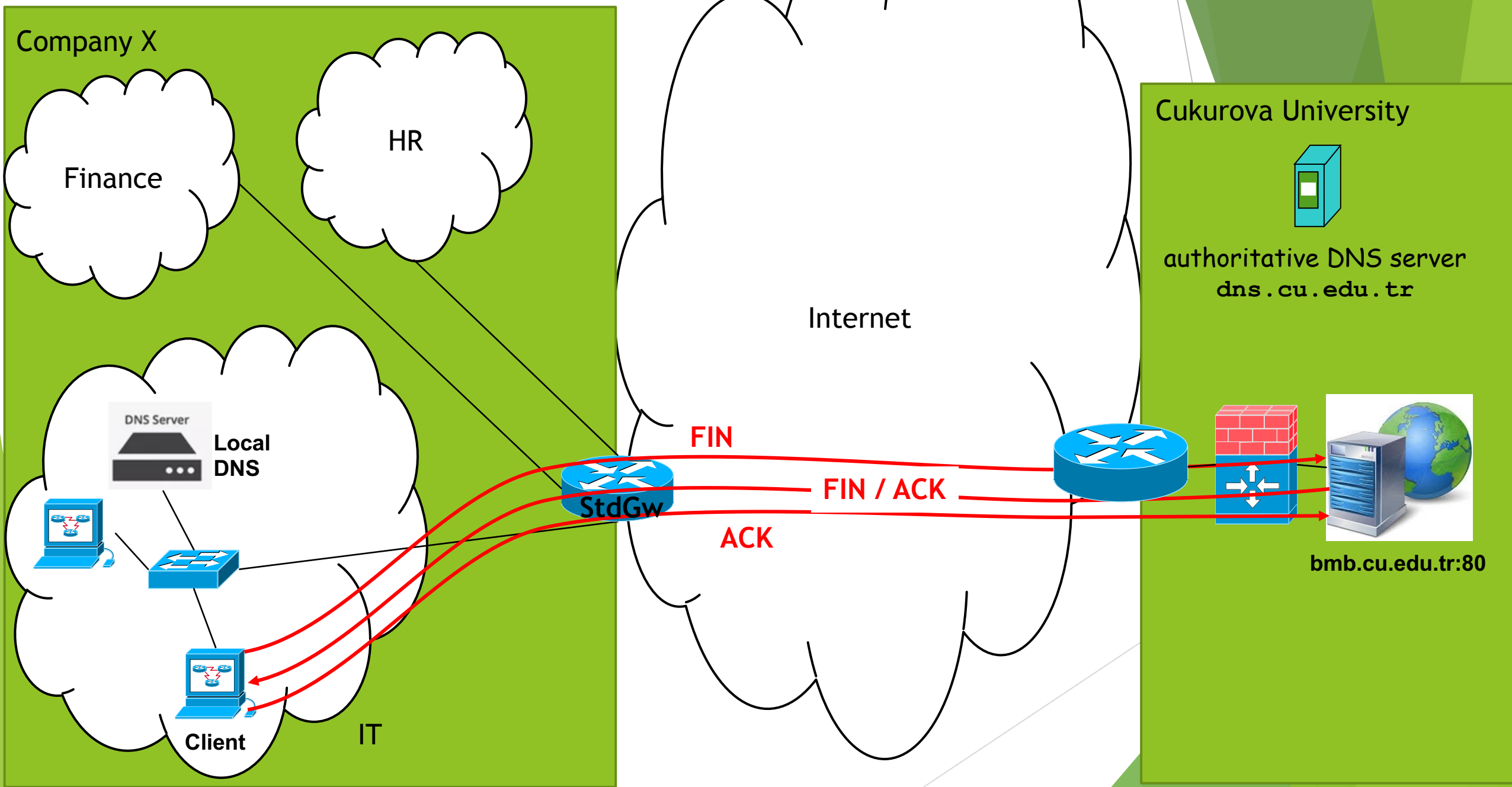
Step 13: Client establishes a TCP connection to the IP of bmb.cu.edu.tr over the StdGw.

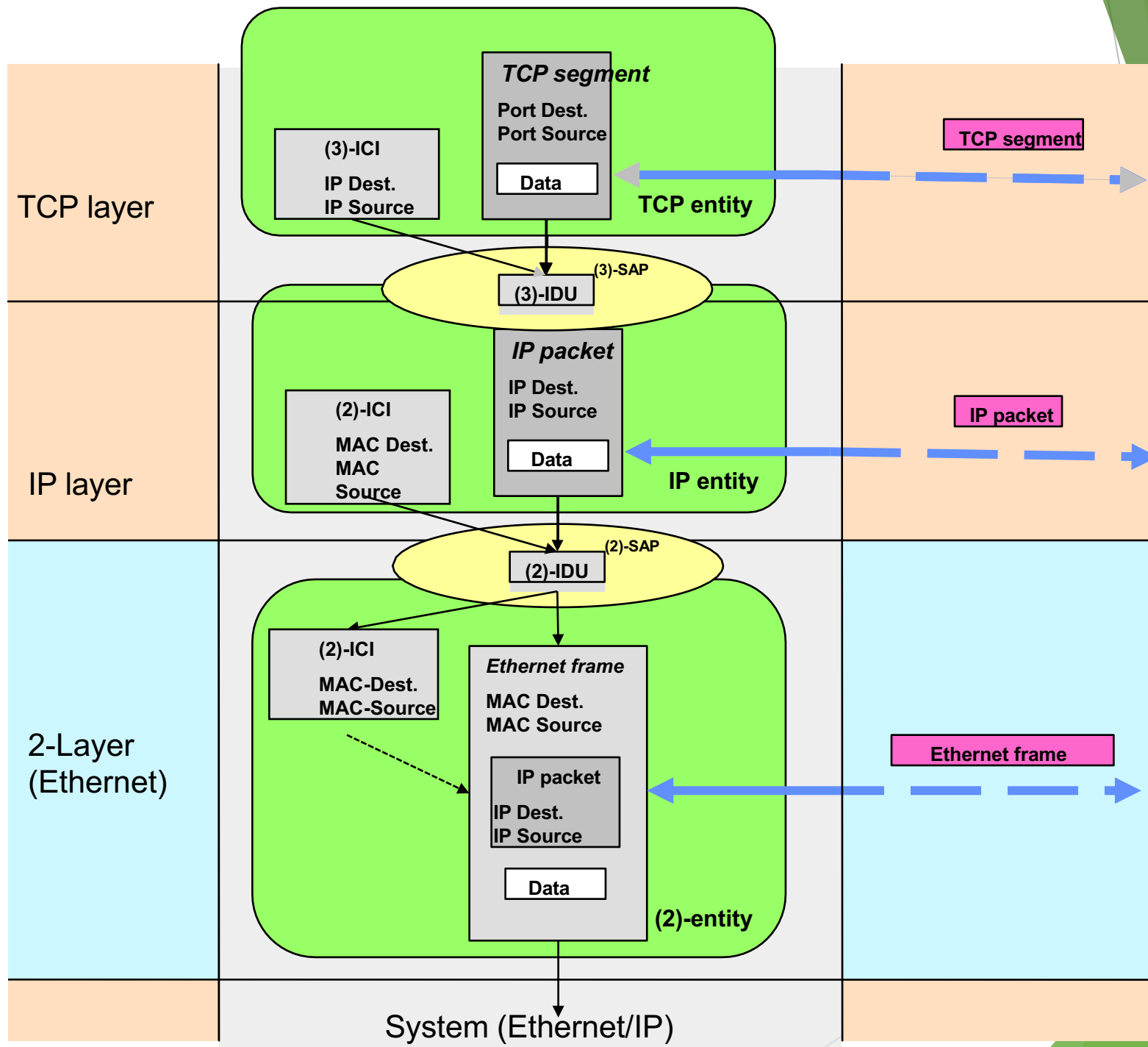


Step 14: Client sends a HTTP GET request to the IP of bmb.cu.edu.tr with URL index.html over the established TCP connection and receives the response.



Step 15: Client and server terminate the TCP connection.





Overview of Application Layer Protocols

